

Use It or Lose It?

Programmable money, negative rates and the design of the digital euro

Speaker script · 20 minutes + 10 minutes Q&A

THE STORY IN TEN SENTENCES

- 1 Use it or lose it?
- 2 Monetary policy stalls when money is hoarded and rates cannot fall below zero.
- 3 Vouchers with a deadline get money spent, and a programmable digital euro could become one.
- 4 But an expiry date pushes out more spending, ends in a fire sale and splits money into vintages.
- 5 Continuous decay keeps every euro equal: it is simply a negative rate that reaches every wallet.
- 6 Yet the digital euro is set at zero, which makes it cash without storage costs and raises the lower bound.
- 7 A hard holding limit stops runs in a crisis, but blocks ordinary users in normal times.
- 8 When costs have a cliff and benefits do not, price and quantity together beat either alone.
- 9 The optimal digital euro has three numbers: a free tier, a Pigouvian rate above it, and a cap.
- 10 The digital euro should bear interest: zero for payments, decay above, and a cap that moves with the cycle.

TIMING

1	Title	0:30
2	The problem	2:00
3	Vouchers	1:45
4	Expiry	2:30
5	Decay	2:15
6	Zero rate	2:30
7	Holding limit	2:00
8	Price vs quantity	2:30
9	Design	3:00
10	Conclusion	1:00
		20:00

Checkpoints: slide 4 done by 6:45, slide 6 done by 11:30, slide 8 done by 16:00.

NUMBERS TO KNOW

- Eurosystem balance sheet peak: €8.8tn (June 2022). Inflation 2014–16: 0.3%.
- Deposit facility rate ≤ 0 from 2014 to 2022; lowest -0.50% .
- China coupons: ¥3 per ¥1. France: MPC 61% vs 23%.
- Expiry vs decay, 30 days: 84% vs 73% useful spending; 16% dumped.
- Art. 16(8): “shall not bear interest.”
- Crisis outflow: €699bn at €3,000 (ECB); €4.7tn with no cap.
- Design: free tier €510, rate -2.4% , cap €3,000.
- MRO 2.65%; household overnight deposits 0.26%.

SLIDE 1

0:30

Use It or Lose It?

Programmable money, negative rates and the design of the digital euro

[Presenter name]

October 2026

SAY

- “Give people money with a deadline, and they spend it. Today I ask what happens if a central bank builds that idea into money itself, and what it means for the digital euro.”

SLIDE 2

2:00

THE PROBLEM

Monetary policy stalls when money is hoarded and rates cannot fall below zero

MONEY IS NOT SPENT

€8.8tn

Eurosystem balance sheet at its peak, June 2022

yet inflation averaged 0.3% in 2014–16:
new money was held, not spent

RATES CANNOT FALL

8 years

with the deposit facility rate at or below zero, 2014–22

floor at –0.50%: below that, savers switch to cash, which pays zero

$$MV \downarrow = PY$$

$$i \geq -c_{\text{cash}}$$

Use It or Lose It?

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SAY

- Two things went wrong with monetary policy in the last decade.
- First, money was created but not spent. The Eurosystem balance sheet reached €8.8 trillion, yet inflation averaged 0.3% in 2014–16. Velocity fell.
- Second, rates could not go far below zero. For eight years the deposit rate was at or below zero, but never below –0.50%, because savers can always switch to cash, which pays zero.
- “So: money is not spent, and rates cannot fall. Keep both in mind; I will come back to the second one.”

IF ASKED

Q Inflation is now above target. Why does this matter?

A The lower bound bound for most of 2009–2021. The digital-euro regulation will last decades and should work in both regimes.

SLIDE 3

1:45

A TEMPTING FIX

Vouchers with a deadline get money spent, and a programmable digital euro could become one

CHINA, DIGITAL COUPONS, 2020

¥3

extra spending per ¥1 of subsidy

FRANCE, RANDOMISED PREPAID CARDS

61% vs 23%

one-month MPC with vs without a three-week expiry

A digital euro is programmable. Could money itself carry a deadline?

Xing, Zou, Yin, Wang & Li (AEJ: Macro 2023); Ding, Jiang, Msall & Notowidigdo (2024); Boehm, Fize & Jaravel (2024).

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SAY

- Governments found a way around the first problem: vouchers with a deadline.
- In 2020, Chinese cities gave out digital coupons, valid for about a week. Each yuan of subsidy produced about three yuan of extra spending.
- A randomised trial in France: with a three-week expiry, 61% of the transfer was spent within a month; without, 23%.
- “A digital euro is programmable. So why not give money itself a deadline?”

IF ASKED

Q Weren't the Chinese results driven by minimum-spend thresholds?

A Partly, as Ding et al. show. The French trial isolates the deadline.

SLIDE 4

2:30

THE FLAW

But an expiry date pushes out more spending, ends in a fire sale and splits money into vintages

MODEL

A euro is worth V to its holder. A good purchase is worth 1, a fire sale $\phi < 1$. Searching at speed κ costs $\kappa^2/2a$, so

$$\kappa^* = a(1 - V)$$

84% vs **73%**

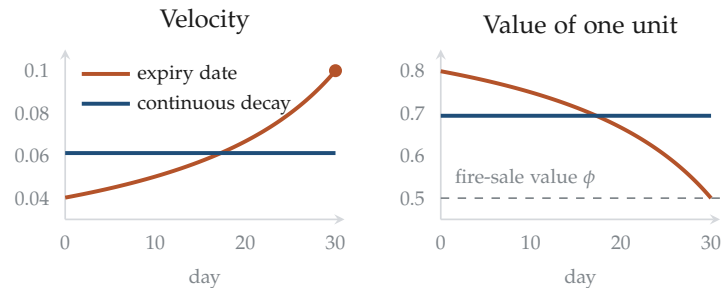
useful spending within 30 days

16%

of issuance dumped on the last day

Both schemes calibrated to the same average velocity over 30 days; $a = 0.2$, $\phi = 0.5$, $r = 4\%$.

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0.80 → 0.50

value of one euro, by its age

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SAY

- To see what an expiry date does, take a holder who must search for good uses of money. Searching faster is costly. A bad use, junk or a discounted sale, is always available.
- The key result: the less money is worth holding, the faster it is spent.
- Compare an expiry date with continuous decay, tuned to the same average velocity over 30 days.
- Expiry actually gets more useful spending done: 84% against 73%. It forces the whole balance out.
- But look at the last day: 16% of issuance is still unspent and goes to junk. And on the right, an expiring euro is worth 0.8 when new and 0.5 when old. One euro is no longer one euro.

IF ASKED

Q Why a quadratic search cost?

A It is the simplest convex cost. With a linear cost, velocity would literally explode at the deadline; convex is the conservative case.

Q Why match average velocity?

A Velocity is the target in a liquidity trap. Matching total spending is impossible here, because decay always burns part of the balance.

SLIDE 5

2:15

THE REPAIR

Continuous decay keeps every euro equal: it is simply a negative rate that reaches every wallet

	Expiry	Decay
Fire sale at the deadline	3–55%	none
Value of one unit	by age	equal
One euro = one euro	no	yes
Applies to all money	no	yes
Is it an interest rate?	no	$i = -\lambda$

VELOCITY UNDER DECAY

$$\kappa^* \approx \sqrt{2a(r + \lambda)}$$

the square-root rule of Baumol–Tobin money demand

Decay λ	1%	2%	3%
Velocity	+12%	+22%	+32%

It unblocks both channels: money moves faster, and the rate reaches retail balances.

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SAY

- Decay has none of these problems: no fire sale, every unit worth the same, and it can apply to all money.
- And decay at rate λ is exactly an interest rate of $-\lambda$.
- Velocity responds with the square root of the holding cost, the same rule as Baumol–Tobin money demand. One percent of decay raises velocity by about 12%.
- “So decay unblocks both channels from slide 2: money moves faster, and a negative rate finally reaches retail balances.”

IF ASKED

Q Couldn't a smart contract track the vintages of expiring money?

A Yes, but then shops must take old and new euros either at par, which invites arbitrage, or at different prices, which ends the singleness of money.

SLIDE 6

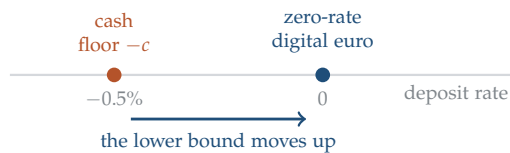
2:30

THE TWIST

Yet the digital euro is set at zero, which makes it cash without storage costs and raises the lower bound

“The digital euro shall not bear interest.”

Draft regulation on the digital euro, Art. 16(8)



Banks cannot pay less than the best central-bank alternative:

$$r_D \geq \max_j \{ i_j - c_j \}$$

Cash: $0 - c < 0$. Digital euro: $0 - 0 = 0$.

The lower bound and disintermediation are one problem: a central-bank rival to deposits.

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SAY

- But the draft regulation goes the other way. Art. 16(8): the digital euro shall not bear interest. Not positive, not negative.
- Why does that matter? A bank cannot pay depositors less than their best central-bank alternative, net of holding costs.
- Cash pays zero but is costly to store and insure, so the floor sits a bit below zero. That is why -0.50% was possible.
- A zero-rate digital euro is cash without those costs. It pushes the floor back up to exactly zero.
- “The lower bound and disintermediation are the same problem: a central-bank rival to bank deposits.”

IF ASKED

Q Retail deposit rates never went negative anyway.

A For small balances banks chose not to charge. Large and corporate deposits did pay negative rates. A zero-rate digital euro makes zero a legal floor for everyone, up to the limit.

THE TRADE-OFF

A hard holding limit stops runs in a crisis, but blocks ordinary users in normal times

NORMAL TIMES: MONEY FOR PAYMENTS

18%

of people want to hold more than €3,000

A cap blocks them. A price would only make them pay.

⇒ price is enough

CRISIS: FLIGHT TO SAFETY

€4.7tn

deposit outflow without a cap (€0.65tn with €3,000)

A rate gap of 2% means nothing next to the fear of losing a deposit.

⇒ only quantity works

Survey demand: Lambert et al. (ECB WP 2980, 2024). Outflows: own calibration to ECB technical data on holding limits (Oct 2025).

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SAY

- The ECB's answer to the disintermediation risk is a hard cap, around €3,000.
- In normal times people hold the digital euro to pay. 18% would like to hold more than €3,000; the cap simply blocks them. A price would let them hold more if they pay for it.
- In a crisis people run for safety. Without a cap, my calibration gives an outflow of €4.7 trillion; with €3,000 it is about €0.65 trillion. A rate gap of two percent means nothing next to the fear of losing a deposit.
- "So price works in normal times, and only quantity works in a crisis."

IF ASKED

Q *Where do the outflow numbers come from?*

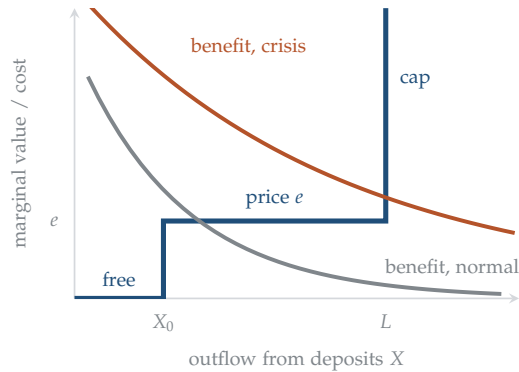
A The ECB published €156bn at a €500 cap and €699bn at €3,000. I fit the survey distribution of deposits to those two points; the implied population is 360 million, the actual euro-area population. Without a cap, the model gives €4.7tn, close to Meller and Soons' "almost €5 trillion".

SLIDE 8

2:30

THE PRINCIPLE

When costs have a cliff and benefits do not, price and quantity together beat either alone



WEITZMAN (1974)

Advantage of prices over quantities:

$$\Delta = \frac{\sigma^2 (b - c)}{2 b^2}$$

b : slope of benefit, c : slope of cost

- Flat cost, steep benefit: **price**
- Vertical cost at L : **quantity**
- Both on one curve: **tiered rate with a cap** (Roberts & Spence 1976)

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SAY

- This is an old question in economics: prices or quantities? Weitzman's answer depends on slopes.
- If the benefit curve is steeper than the cost curve, use a price. If the cost curve is steeper, use a quantity.
- Here the cost of deposit outflows has three segments: free while banks absorb it, a constant cost e while they replace funding, and a cliff once liquidity buffers are gone.
- In normal times the benefit curve crosses the flat part: price. In a crisis it shifts out and hits the cliff: quantity.
- "Both happen on one curve, so the answer is both: a tiered rate with a cap, as Roberts and Spence showed."

IF ASKED

Q Isn't the ECB already proposing tiering?

A Bindseil (2020) proposed a two-tier rate. What I add is the reason, from the shape of the cost curve, and a calibration of each tier.

SLIDE 9

3:00

THE DESIGN

The optimal digital euro has three numbers: a free tier, a Pigouvian rate above it, and a cap

FREE TIER M_1

€510

$$N\mathbb{E}[\min(t, M_1)] = X_0$$

payments use only what banks
absorb for free (€127bn)

RATE ABOVE THE TIER

−2.4%

$$-e = -(\text{MRO} - r_D)$$

holders pay the bank's cost of
replacing the deposit

CAP $\bar{M}$

€3,000

$$N\mathbb{E}[\min(d, \bar{M})] = \theta D$$

crisis outflow stays within bank
liquidity tolerance

	Users blocked	Outflow if rates < 0	Outflow in a crisis
Cap only (0%, €3,000)	18.5%	€648bn	€648bn
Price only (tier, −2.4%)	0%	€158bn	€4,716bn
Hybrid (tier, −2.4%, cap)	7.3%	€158bn	€648bn

Calibrated to ECB WP 2980 survey distributions and ECB holding-limit outflows (Oct 2025); MRO 2.65%, household overnight rate 0.26%.

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SAY

- Each segment of the cost curve gives one number.
- Free tier: payments should only use what banks absorb at no cost. The ECB expects digitalisation to bring €127bn into deposits; spending exactly that gives a free tier of about €510.
- Above the tier, the rate is minus the bank's cost of replacing deposit funding: the refinancing rate minus the deposit rate, about −2.4%.
- The cap keeps crisis outflows within what banks can bear; at today's tolerance that is €3,000.
- The table: the cap alone blocks 18% of users and leaks €648bn when rates are negative. Price alone fails in a crisis: €4.7 trillion. The hybrid blocks 7%, leaks €158bn, and holds in a crisis.

IF ASKED

Q Is −2.4% not a huge penalty?

A In normal times almost nobody pays it; it only stops the digital euro from becoming a savings account. And it is an upper bound: if central-bank funding replaces deposits cheaply, the social cost is lower.

Q Why €127bn for the free tier?

A It leaves banks no worse off than today. The sensitivity: €100bn to €300bn gives a tier between €370 and €1,840 (backup).

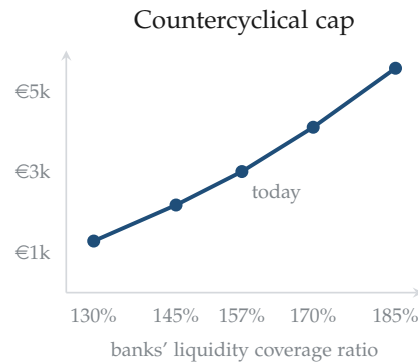
SLIDE 10

1:00

CONCLUSION

The digital euro should bear interest: zero for payments, decay above, and a cap that moves with the cycle

1. Deadlines get money spent, but expiry belongs to a voucher, not to money
2. Decay keeps every euro equal and is simply a negative rate
3. A zero-rate digital euro is costless cash: it raises the lower bound
4. Free tier, Pigouvian decay, and a cap tied to bank liquidity



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SAY

- Read the four points.
- On the chart: the cap depends on how much liquidity banks have. When buffers are thick, the cap can rise; when they thin, it should fall. A countercyclical cap.
- Close: “Deadlines belong to vouchers. Money can only carry a rate. The digital euro should be allowed to carry one: zero for payments, decay above, and a cap that moves with the cycle.”

MORE QUESTIONS

Q *Isn't negative interest on CBDC already in Bordo and Levin (2017)?*

A Yes, as an idea. The new parts here are why it must be decay rather than expiry, and how a rate and a cap should be combined and calibrated.

Q *Would people accept a digital euro that can shrink?*

A Balances up to the free tier never shrink. Above it, the charge only makes the digital euro no better than a bank account.

Q *The ECB says the digital euro will not be programmable money.*

A That means no restrictions on what it buys. Decay restricts nothing; it is a rate on a balance.

Q *Doesn't a countercyclical cap signal panic when it is lowered?*

A It should move by rule with published liquidity data, not by discretion, so that it is anticipated rather than read as a signal.

Q *Your model is partial equilibrium.*

A Yes. The qualitative results do not depend on general equilibrium; the magnitudes are calibrated to the ECB's own numbers.

GLOSSARY

Velocity	Spending per unit of money per period, $V = PY/M$.
MPC	Share of an extra euro spent within a period.
Lower bound (ZLB / ELB)	The rate below which savers switch to cash; slightly below zero because cash is costly to hold.
Demurrage / decay	A continuous charge on holding money; a negative interest rate.
Singleness of money	Every form of a currency exchanges one-for-one at par.
Baumol–Tobin	Inventory model of money demand; holdings fall with the square root of the holding cost.
Disintermediation	Deposits leaving banks for central-bank money.
Weitzman (1974)	Prices beat quantities when benefits are steeper than costs, and vice versa.
Roberts–Spence (1976)	A price schedule with a quantity ceiling beats either instrument alone.
Pigouvian rate	A charge equal to the cost one holder imposes on others.
LCR	Liquidity coverage ratio: liquid assets over 30-day stressed outflows; minimum 100%.
MRO	Main refinancing operations rate, 2.65% since 16 Sep 2026.
Art. 16(8)	Draft regulation: "The digital euro shall not bear interest."